

LIBRARY BOOK REQUEST SYSTEM

1. INTRODUCTION

The Library Book Request System is a ServiceNow-based application developed to automate the process of book requests in a college library. Students can request books online, and the librarian receives notifications for each request.

[Insert Image 1: Project Overview / App Homepage]

2. OBJECTIVE

The primary objective of the Library Book Request System is to automate and simplify the process of requesting books from the college library using the ServiceNow platform. The system provides an online interface where students can submit book requests by entering details such as Student Name, Student ID, Book Name, Author Name, and Issue Date. This eliminates manual paperwork, improves data accuracy, and enables efficient management of library requests.

The project also aims to enhance the library workflow through automation and validation features. A unique Request Number is automatically generated for every request, ensuring easy tracking and identification. The system validates that the Issue Date is not in the past and automatically notifies the librarian whenever a new request is submitted. These features help reduce administrative effort, improve communication, and ensure a faster and more reliable book request process.

The main objectives of this project are:

- To digitize library book request process
- To validate issue date (prevent past dates)
- To auto-generate unique request numbers
- To notify librarian on every request submission

[Insert Image 2: Requirement Summary / App Engine Studio Overview]

3. SYSTEM DESIGN

3.1 Application Setup

To create and configure the ServiceNow application environment

Steps

1. Open App Engine Studio

2. Create a new application Application Details:

- Name: Library Book Request
- Scope: x_1877534_librar_0

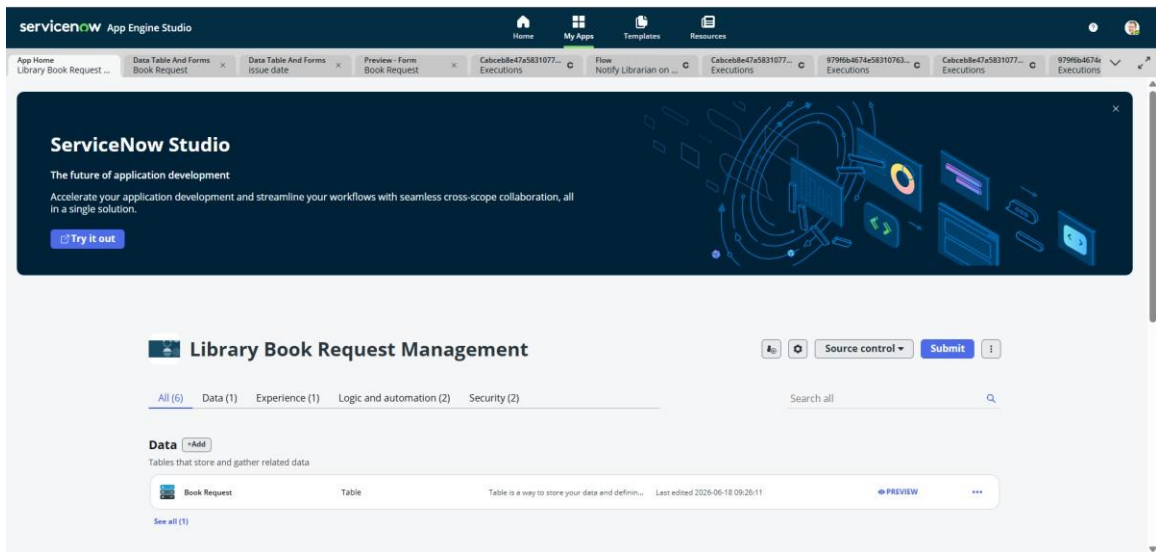
The image shows two screenshots of the ServiceNow Developer portal and App Engine Studio interface.

Top Screenshot: Manage my instance (dev405553)

This screenshot shows the 'Manage my instance' page for instance dev405553. The page includes a status section with a 'Refresh' button, a table of installed agents (Build agent, ServiceNow studio, App engine studio), and a 'Useful links' section with links to the PDI Guide, PDI FAQs, Managing your PDI, and the Developer advocate blog. Below this is a 'Plugins for your instance (45)' section with a search bar and filters for 'All activation statuses' and 'All demo data statuses'.

Bottom Screenshot: My apps

This screenshot shows the 'My apps' page in App Engine Studio. It features a search bar, an 'Import app' button, and a 'Create app' button. Below these are two app cards: 'Library Book Request Man...' (last opened 2026-06-19 18:52:05) and 'Laptop Request Managem...' (last opened 2026-06-19 02:18:14). The page also has a 'Sort (newest)' dropdown.



3.2 Table Design

A custom table named Book Request was created in ServiceNow with fields such as Request Number, Student Name, Student ID, Book Name, Issue Date, Request Date, and Status.

Column label	Column name	Type	Reference	Max length	Default value	Display	Updated
Author Name	author_name	String		100			2026-06-18 09:39:15
Book Name	book_name	String		100			2026-06-18 09:39:15
Created	sys_created_on	Date/Time					2026-06-18 09:26:11
Created by	sys_created_by	String		40			2026-06-18 09:26:11
Issue Date	issue_date	Date					2026-06-18 09:39:14
Number	number	String		40	javascript:global.getNextObj...		2026-06-18 09:26:11
Request Date	request_date	Date					2026-06-18 09:39:14
Return Date	return_date	Date					2026-06-18 09:39:14
Status	status	Choice			Requested		2026-06-18 09:39:13
Student ID	student_id	String		20			2026-06-18 09:39:15
Student Name	student_name	String		100			2026-06-18 09:39:16
Updated	sys_updated_on	Date/Time					2026-06-18 09:26:11
Updated by	sys_updated_by	String		40			2026-06-18 09:26:11

3.3 Auto Number Generation

Auto number generation was implemented to create unique request numbers like B00000001, B00000002, etc.

Book Request
New record View: Default view

Number: 0000000001

Student Name:

Student ID:

Book Name:

Author Name:

Issue Date:

Request Date:

Status: Requested

Return Date:

Submit

3.4 Issue Date Validation

A Client Script was used to ensure the issue date is not in the past. If the selected date is less than today, submission is blocked.

Library Book Request
New record

Issue Date cannot be in the past

Invalid insert

Request Number: LIB0001019

Student Name: Vignesh

Student ID: 2345

Book Name: nun

Author Name: Vicky

Issue Date: 2026-06-22

Return Date: 2026-06-14

Submit

3.5 Notification System

A notification was configured to send an email to the librarian whenever a new book request is created.

Emails View: Outbox

Created

Search

All > Mailbox > Outbox > Created on Today

Q

Created	Recipients	Subject	State
2026-06-19 06:00:09	fred.luddy@example.com	On-Call Schedule: Hardware	Ready
2026-06-19 18:40:39	librarian@college.edu		Ready
2026-06-19 01:21:59	admin@example.com	Request REQ0010001 was approved	Ready
2026-06-19 21:05:48	librarian@college.edu	New Book Request Submitted: \${request_number}	Ready
2026-06-19 01:00:09	alleen.mottorn@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	Ready
2026-06-19 21:08:39	librarian@college.edu		Ready
2026-06-19 20:59:48	librarian@college.edu	New Book Request Submitted: \${request_number}	Ready
2026-06-19 18:40:40	librarian@college.edu	New Book Request Submitted: \${request_number}	Ready
2026-06-19 01:21:59	admin@example.com	Request REQ0010001 was created	Ready
2026-06-19 21:05:49	librarian@college.edu		Ready
2026-06-19 01:00:09	alleen.mottorn@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	Ready
2026-06-19 20:40:29	librarian@college.edu		Ready
2026-06-19 20:59:49	librarian@college.edu		Ready
2026-06-19 00:00:10	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	Ready
2026-06-19 06:00:09	chris.harris@example.com	On-Call Schedule: Nimsoft desk	Ready
2026-06-19 21:08:32	librarian@college.edu	New Book Request Submitted: \${request_number}	Ready
2026-06-19 01:21:59	fred.luddy@example.com	Catalog Task SCTASK0010002 has been assigned to group Software	Ready
2026-06-19 20:36:28	librarian@college.edu	New Book Request Submitted: \${request_number}	Ready
2026-06-19 20:36:30	librarian@college.edu		Ready
2026-06-19 10:00:29	admin@example.com	[TSM Analytics] Daily Data Collection completed with warning	Ready

3.6 Email Output

The system generates an email which is stored in sys_email logs and sent to the librarian.

Email View: Outbox

Mailbox: Outbox

State: Ready

Target: Book Request: BOO0001022

User:

UID: N/A

Created: 2026-06-19 18:40:39

Deleted:

Weight: 0

Importance:

Originating Event and Notification

Event	Notification
notification_engine.process	Book Request - Notify Librarian

Subject:

Recipients: librarian@college.edu

Body:

<html><head><style>

#snc_notification_preference {

text-decoration: none;

}

#snc_notification_unsubscribe {

text-decoration: none;

}

</style>

</head><body><p>Hello Librarian,</p>

<p>A new book request has been submitted.</p>

<p>Request Number:
Student Name: HERSHANA
Student ID: 313
Book Name: PYTHON
Issue Date:
Request Date: 2026-06-22
Status: Requested</p>

<p>Please review and take action.</p>

<p>Thank you.</p>

Unsubscribe | Preference

3.7 Flow Execution

A flow was created to trigger actions when a record is inserted and send notifications accordingly.

[Insert Image 8: Flow Execution]

4. WORKING PROCESS

1. Student submits request
2. Issue date is validated

3. Request number is auto-generated
4. Record is saved
5. Notification is sent to librarian
6. Email is logged in system

[Insert Image 9: Form Submission]

5. OUTPUT

Book requests are successfully created, validated, and notified to the librarian with proper tracking.

Preview Email

A new library book request has been submitted.

Request Number:

Student Name: Vignesh

Student ID: 245

Book Name: nun

Author Name: chris

Issue Date: 2026-06-24

Return Date: 2026-06-26

[Unsubscribe](#) | [Notification Preferences](#)

Ref:MSG0000370_TnzpI4ytUY8zhEXyrkL5

6. CONCLUSION

The system successfully automates library book request management using ServiceNow, reducing manual effort and improving efficiency.